CHETAN DNYANESHWAR KHARKAR

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Authorized to work in the U.S. • Open to Relocate

PROFESSIONAL SUMMARY

SnowPro Core certified Data Engineer with 3+ years of experience building end-to-end data pipelines, cloud data warehouses, and analytics platforms. At Fidelity Investments, built SOX-compliant ETL systems processing 10M+ records, cutting compliance reporting time from hours to seconds and eliminating 20+ hours/week of manual work. Strong across Snowflake, Airflow, PySpark, Kafka, AWS, SQL, and data modeling, with hands-on GenAI and RAG pipeline experience and $50K+ in cost savings identified through utility analytics at WPI.

CERTIFICATIONS

SnowPro Core Certification | Snowflake

PROFESSIONAL EXPERIENCE

Data Engineer | Fidelity Investments **Jul 2025 – May 2026**

Boston, MA — Financial Services (Fortune 100)

- Eliminated 20+ hrs/week of manual SOX compliance work by automating control attestation for 100+ control owners with an Airflow-scheduled Python ETL and a Streamlit app for report distribution, response tracking, audit evidence generation, and automated reminders.
- Reduced compliance reporting time from hours to seconds by building Airflow ETL pipelines ingesting 10M+ Oracle records into Snowflake via AWS S3, replacing manual report preparation with automated daily refreshes feeding Tableau dashboards.
- Automated a biweekly portfolio cash-monitoring control with a fault-tolerant Python ETL pipeline, using retries with backoff, record-level quarantine, validation gates, and failure alerting, saving 10+ hrs per reporting cycle.
- Implemented governance controls within Fidelity's SOX framework across pipelines: role-based access in Snowflake, row-count/null/schema validation gates, column-level documentation, and end-to-end lineage from Oracle through S3 and Snowflake to Tableau.
- Lowered Snowflake compute costs by right-sizing warehouses per workload, enabling auto-suspend for reporting warehouses, and building materialized views and summary tables for repeated dashboard aggregations, while meeting reporting SLAs.
- Owned production pipeline reliability in on-call rotation, triaging and resolving incidents within SLA using PagerDuty and Snowflake Query Profile.

Data Engineer | Worcester Polytechnic Institute **Nov 2024 – Jul 2025**

- Designed a Snowflake star schema warehouse for facilities and utility analytics, separating transaction facts from item, location, department, supplier, and date dimensions, improving reporting query performance by 40% over the prior flat structure.
- Surfaced $50K+ in cost-saving opportunities by building statistical anomaly detection in Python, comparing building-level utility usage and cost against historical and seasonal baselines and flagging outliers for facilities review.
- Built Python ingestion pipelines consolidating utility and facilities data from multiple source systems into staged and curated Snowflake layers, with row-count, duplicate, and required-field validation, reducing manual reconciliation.
- Partnered with facilities and IT teams to define KPIs, consumption by building, cost trends, and anomaly flags, then built Tableau and Power BI dashboards enabling self-service analytics for non-technical staff.
- Established data governance in Snowflake through schema versioning, column-level documentation, and role-based access control, improving data trust and auditability across the team.

Data Engineer | Vodafone Idea (VI) **Dec 2022 – Sep 2024**

- Built and maintained ETL pipelines processing millions of telecom subscriber and network performance records daily, supporting batch and near-real-time churn and network KPI reporting for BI and analytics teams.
- Contributed to migrating legacy on-prem Oracle ETL workflows to Azure Data Factory pipelines, validating source-to-target parity and cutting pipeline execution time by 30% with improved scheduling, retries, and monitoring.
- Optimized SQL stored procedures and reporting queries through early filtering, staged transformations, and pre-aggregated reporting tables, reducing scan volume and dashboard load times on high-volume telecom data.
- Reduced data incidents reaching downstream reports by 40% with automated Python quality monitoring: row-count, null, duplicate, schema-change, and feed-drop checks with alerting.

KEY PROJECTS

Compensation Analytics Platform | Boston Scientific (Sponsored Capstone) **Jan 2026 – May 2026**

- Led data modeling to replace a fragmented 1M+ row Excel compensation model with a Snowflake star schema centralizing sales performance data across rep, job role, product, territory, and time dimensions, enabling budget scenario testing and quarter-end compensation forecasting that wasn't feasible in Excel.
- Modeled a rules-as-data schema storing comp plan logic (tiers, accelerators, bonuses) as versioned data by plan and year, powering a two-phase Snowflake stored-procedure engine that matches rep performance to rules and computes payouts, enabling plan changes without rewriting formulas.
- Built the Streamlit app for comp rule entry and scenario testing, and integrated Snowflake Cortex Analyst with a custom semantic model mapping business terms (commission, attainment, quota) to the schema, enabling plain-English querying for finance and sales ops.

Real-Time Streaming Data Pipeline **Feb 2026 – Mar 2026**

- Built an event-driven pipeline streaming real-time gaze-tracking events through Kafka into PySpark Structured Streaming, evaluating each event against detected screen regions and triggering interventions with sub-second end-to-end latency at ~60 events/sec.
- Containerized the system with Docker Compose and implemented JSON schema validation, Kafka offset-based recovery, and decoupled producer and alert topics for fault-tolerant, low-latency delivery.

Clinical NLP RAG Pipeline **Feb 2025 – Apr 2025**

- Built a retrieval-augmented generation pipeline with LangChain, BioBERT embeddings, and FAISS vector search for clinical text, achieving 0.91 precision on medical entity extraction evaluated against a labeled dataset.
- Engineered document chunking, embedding, and vector retrieval workflows to surface relevant clinical passages for grounded generation, tuning retrieval parameters to improve precision on domain-specific medical text.

TECHNICAL SKILLS

Data Engineering: Apache Airflow, Apache Kafka, PySpark, dbt, ETL/ELT, Streaming Pipelines, Docker, CI/CD (GitHub Actions)
Data Warehousing: Snowflake, Snowpark, Star Schema Design, Data Lakes, Data Modeling
Cloud Platforms: AWS (S3, EC2), Azure (Data Factory)
Programming: Python, SQL, Shell Scripting
AI / GenAI: LangChain, FAISS, RAG Pipelines, NLP, Snowflake Cortex AI
Visualization & Apps: Tableau, Power BI, Streamlit
Domain Expertise: SOX Compliance, Data Governance, Portfolio Analytics, Financial Data

EDUCATION

M.S. Data Science, GPA: 3.9/4.0 | Worcester Polytechnic Institute **2024 – 2026**

B.E. Electronics & Telecommunication, GPA: 9.1/10 | University of Pune **2019 – 2023**

ACHIEVEMENTS & RECOGNITION

- Employee of the Month, Fidelity Investments: Recognized for automating SOX control attestation and compliance reporting workflows.
- Top 2% Globally, Jane Street Kaggle Competition: Ranked in the top 2% out of 10,000+ participants worldwide in a competitive quantitative data science challenge.